Research Statement

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I develop trustworthy AI systems for science, medicine, and human-machine interaction. **My research asks how AI can reason from real-world evidence, collaborate effectively with people, and operate reliably in high-stakes settings.** I pursue this goal through three connected directions. First, I build foundation models for scientific and clinical reasoning and design human-AI collaboration frameworks that enable their reliable use in real-world workflows. Second, I develop autonomous AI systems that integrate evidence, memory, and tools to reason and act effectively over long-context multimodal understanding while reducing computational cost. Third, I design EEG-based brain-computer interface (BCI) that analyzes brain dynamics to enable accurate and responsive human-machine interaction. Together, these directions advance my long-term goal of creating AI systems that reliably support scientific discovery, clinical decision-making, and human-centered intelligent technologies.

1 Trustworthy foundation model for science and medicine

Foundation models have the potential to transform science and medicine by enabling human-AI collaboration in real-world workflows. However, current AI systems remain limited by hallucinations and insufficient domain specialization, restricting their deployment in high-stakes settings. Frontier models, in particular, struggle to interpret complex medical imaging data, such as magnetic resonance imaging (MRI) and computed tomography (CT), because they are primarily optimized for general-purpose tasks. To address these challenges, my research advances scalable learning frameworks for large language models (LLMs) and large vision-language models (LVLMs) that integrate domain-specific knowledge, real-world evidence, and human-AI collaboration frameworks to enable reliable and specialized AI systems. **As of August 2026, my open-source models have been collectively downloaded more than 730,000 times on Hugging Face, with over 35,000 downloads per month [1], while my projects have collectively received more than 2,100 GitHub stars and 300 forks [2].** In [3–5], I introduced PodGPT, a continual pre-trained LLM enhanced with informative, up-to-date multilingual podcasts spanning science, technology, engineering, mathematics, and medicine (STEMM), together with a retrieval-augmented generation (RAG) framework that grounds model responses in emerging scientific literature. In [6], I developed ReMIND (Radiology-encoded Multimodal Interpretation for Neurological Disorders), a vision-language learning framework for comprehensive multi-sequence and multi-volumetric brain MRI analysis. ReMIND combines large-scale clinically grounded video instruction tuning with targeted supervised fine-tuning for radiology report generation, enabling clinically validated specialized intelligence for high-stakes medical imaging. **The platform has been adopted by hundreds of researchers, physicians, radiologists, and neurologists across more than 60 cities in Asia, Europe, and North America [7].** I also introduced a model merging framework for multitask neuroimaging interpretation, providing a foundation for personalized medical intelligence systems [8]. I further extended this research to image generation by developing a novel methodology of synthesizing high-resolution 3D lung CT images from textual prompts and anatomical priors [9]. **This work was published in the prestigious *IEEE Transactions on Medical Imaging (IEEE TMI)*, with over 90 citations and an impact factor above 12**. Additionally, I also developed vision models for image quality assessment, further expanding the applicability of AI [10, 11]. My work represents a significant step toward translating trustworthy foundation models into real-world deployable AI systems that support precision science and medicine.

2 Grounded and cost-efficient autonomous AI system

Autonomous AI systems are rapidly emerging as a foundational technology for manufacturing and digital infrastructure, with the potential to drive future economic growth and national security. Despite their remarkable benchmark performance, frontier models remain limited in solving complex real-world tasks due to insufficient multimodal reasoning and increasing cost from extended chain-of-thought reasoning. My research addresses these challenges by designing grounded and cost-efficient autonomous AI systems that integrate memory, tool harness, and evidence grounding to reason and act effectively over long-context, multimodal information while reducing computational cost. In work **published in *Empirical Methods in Natural Language Processing (EMNLP)*, a top-tier venue in AI**, I introduced a multi-agent framework with a task-agnostic memory module inspired by human reading strategies for long-context multimodal understanding [12]. Compared with optical character recognition (OCR)-based approaches, the framework achieved significant performance **improvements of 21.70% and 12.40% over `GPT-4o` and `Claude 3.5 Sonnet`**, respectively, while reducing token consumption and inference cost. Extending this research, in [13], I developed a unified open-source agentic RAG framework that integrates document retrieval, reranking, memory augmentation, evidence grounding, and diagnosis generation into a coherent multi-step reasoning pipeline for both open-ended and closed-form medical question-answering (QA). I further introduced a cache-and-prune memory bank mechanism that efficiently retains relevant evidence over long reasoning horizons, improving both diagnostic accuracy and computational efficiency on challenging medical QA tasks. Across five major benchmarks, the framework achieved superior or competitive performance compared with standalone LLMs. In particular, it achieved a **4.57% improvement in accuracy over `GPT-4`** on the United States medical licensing examination (USMLE) Step 2. Collectively, my research demonstrates the effectiveness of evidence-grounded autonomous agents for long-context, multimodal understanding and high-stakes clinical reasoning.

3 Accurate and responsive EEG-based BCI system

EEG-based BCI offers a non-invasive, portable, and high-temporal-resolution approach for direct human-machine interaction, with the potential to assist individuals with paralysis, stroke, and other severe motor impairments. In my early work, I developed an EEG decoding framework that extracts Morlet wavelet features over the motor cortex and subsequently applied a convolutional neural network for neural decoding [14]. **Published in the *Journal of Neural Engineering*, a leading journal in BCI research with an impact factor above 5, this work has received more than 170 citations, 240 GitHub stars, and 50 GitHub forks [15].** However, practical EEG-based BCIs remain challenged by substantial inter-subject variability and high response latency. Moreover, traditional methods often overlook the functional topology of brain networks, limiting their ability to capture complex neural interactions. To address these challenges, I developed a pioneering graph-based deep learning framework that explicitly models functional relationships among EEG electrodes for robust neural decoding of motor imagery (MI) tasks [16]. **Published in the prestigious *IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)*, a top-tier machine learning journal with an impact factor above 14, this work has received more than 260 citations.** Building on this foundation, I proposed a highly accurate and responsive MI recognition framework capable of decoding scalp EEG recordings **as short as 0.4 seconds** [17]. **This work has received over 100 citations.** My approach demonstrated strong robustness to inter-trial and inter-subject variability while maintaining low-latency inference, representing an important step toward practical, real-time EEG-based BCI applications. To accelerate research in this area, I also open-sourced a deep learning library `EEG-DL` for EEG signal classification that has become a ***de facto* standard for EEG deep learning, receiving more than 1,200**

GitHub stars and 230 GitHub forks and supporting both academic research and industrial applications [18]. Together, these efforts advance the translation of EEG-based MI recognition from laboratory research to practical, real-world BCI systems.

Future work

AI will become trustworthy not simply by scaling model size, but by grounding intelligence in domain knowledge, real-world evidence, memory, and interaction with humans and the environment. Over the next five years, I aspire to advance the foundation of safe, reliable, extensible, and cost-efficient AI systems that can deliver measurable real-world impact across science, medicine, healthcare, education, robotics, and human-machine interaction.

Aim 1: Unified medical imaging foundation model

Could frontier AI interpret complex medical imaging, such as a PET scan? My previous work established the foundation for brain MRI analysis. However, existing models remain largely domain-specific and are limited in their ability to jointly reason across heterogeneous clinical information, including family medical history, laboratory tests, medications, neuropsychological assessments, genetic and molecular biomarkers, and medical imaging, such as MRI, CT, X-ray, whole-slide imaging (WSI), and positron emission tomography (PET). My future research will leverage large-scale multimodal data from diverse cohorts to develop a scalable omnimodal foundation model that learns unified representations across heterogeneous clinical modalities. By integrating imaging, physiological, and clinical information within a shared framework, this model will support a broad spectrum of downstream tasks, including question answering, report generation, diagnosis, lesion localization, and tumor segmentation.

Aim 2: Clinical foundation model for disease progression and personalized care

Could frontier AI interpret and reason over patient's longitudinal trajectory? Existing clinical foundation models are developed under isolated patient encounters, limiting their ability to capture disease progression within an evolving clinical context. My prior research established the foundation for long-context, multimodal understanding. My future research will develop a longitudinal foundation model that learns from event-level healthcare trajectories spanning multiple visits and evolving patient's imaging. This framework will model temporal dependencies across events to enable accurate outcome prediction and personalized decision-making, including diagnosis, prognosis, treatment planning, disease management, and medication recommendation, while remaining robust to missing and evolving patient information.

Aim 3: EEG foundation model for real-time brain activity analysis

Could frontier AI effectively decode brain activity in real time? My prior work significantly improved the accuracy and responsiveness of EEG models. However, translating these algorithmic advances into practical BCI systems requires models that can continuously interpret neural signals while remaining robust to inter- and intra-subject variability and operating under low latency. My future research will develop a foundation model for real-time brain activity prediction by learning generalizable representations from large-scale, heterogeneous EEG data. I will further investigate hardware-software co-design by jointly optimizing model architectures, EEG sensing hardware, and edge computing platforms for real-time inference.

References

[1] **Shuyue Jia**. Hugging Face model repository. `https://huggingface.co/shuyuej/models?sort=downloads`, 2026.

[2] **Shuyue Jia**. GitHub open-source repositories. `https://github.com/SuperBruceJia`, 2026.

[3] **Shuyue Jia**, Subhrangshu Bit, Edward Searls, Lindsey A. Claus, Pengrui Fan, Varuna H. Jasodanand, Meagan V. Lauber, Divya Veerapaneni, William M. Wang, Rhoda Au, et al. MedPodGPT: A multilingual audio-augmented large language model for medical research and education. *medRxiv*, 2024.

[4] **Shuyue Jia**, Subhrangshu Bit, Edward Searls, Meagan V. Lauber, Pengrui Fan, William M. Wang, Lindsey A. Claus, Varuna H. Jasodanand, Divya Veerapaneni, Rhoda Au, et al. PodGPT: An audio-augmented large language model for research and education. *Nature npj Biomedical Innovations*, 2(1):26, 2025.

[5] Meysam Ahangaran, **Shuyue Jia**, Shlok Chitalia, Ambarish Athavale, Jean M. Francis, Michael William O'Donnell, Santhoshi Rupa Bavi, Uma Datta Gupta, and Vijaya B. Kolachalama. Performance of open-source large language models on nephrology self-assessment program. *medRxiv*, 2026.

[6] Diala Lteif, **Shuyue Jia**, Subhrangshu Bit, Artem Kaliaev, Asim Z. Mian, Juan E. Small, Balamurugan Mangaleswaran, Bryan A. Plummer, Sarah A. Bargal, Rhoda Au, et al. Vision-language framework for multi-sequence brain magnetic resonance imaging. *medRxiv*, 2026.

[7] Kolachalama Laboratory. ReMIND: Frontier vision-language AI for neuroimaging. `https://remind.bu.edu/`, 2026.

[8] Subhrangshu Bit, Osman B. Guney, **Shuyue Jia**, and Vijaya B. Kolachalama. Gradient-guided adapter merging for neuroimaging vision-language models. *medRxiv*, 2026.

[9] Yanwu Xu, Li Sun, Wei Peng, **Shuyue Jia**, Katelyn Morrison, Adam Perer, Afrooz Zandifar, Shyam Visweswaran, Motahhare Eslami, and Kayhan Batmanghelich. MedSyn: Text-guided anatomy-aware synthesis of high-fidelity 3D CT images. *IEEE Transactions on Medical Imaging*, 43(10):3648–3660, 2024.

[10] **Shuyue Jia**, Baoliang Chen, Dingquan Li, and Shiqi Wang. No-reference image quality assessment via non-local dependency modeling. In *24th IEEE International Workshop on Multimedia Signal Processing, MMSP 2022, Shanghai, China, September 26-28, 2022*, pages 1–6. IEEE, 2022.

[11] Zhaopeng Feng, Keyang Zhang, **Shuyue Jia**, Baoliang Chen, and Shiqi Wang. Learning from mixed datasets: A monotonic image quality assessment model. *Electronics Letters*, 59(3):e12698, 2023.

[12] Li Sun, Liu He, **Shuyue Jia**, Yangfan He, and Chenyu You. DocAgent: An agentic framework for multi-modal long-context document understanding. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, pages 17701–17716. Association for Computational Linguistics, 2025.

[13] **Shuyue Jia**, Subhrangshu Bit, Varuna H. Jasodanand, Yi Liu, and Vijaya B. Kolachalama. Agentic memory-augmented retrieval and evidence grounding for medical question-answering tasks. *International Journal of Medical Informatics*, page 106339, 2026.

[14] Yimin Hou, Lu Zhou, **Shuyue Jia**, and Xiangmin Lun. A novel approach of decoding EEG four-class motor imagery tasks via scout ESI and CNN. *Journal of Neural Engineering*, 17(1):016048, 2020.

[15] **Shuyue Jia**. EEG motor imagery classification via convolutional neural networks. `https://github.com/SuperBruceJia/EEG-Motor-Imagery-Classification-CNNs-TensorFlow`, 2018.

[16] **Shuyue Jia**, Yimin Hou, Xiangmin Lun, Ziqian Hao, Yan Shi, Yang Li, Rui Zeng, and Jinglei Lv. GCNs-Net: A graph convolutional neural network approach for decoding time-resolved EEG motor imagery signals. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6):7312–7323, 2022.

[17] **Shuyue Jia**, Yimin Hou, Xiangmin Lun, Shu Zhang, Tao Chen, Fang Wang, and Jinglei Lv. Deep feature mining via the attention-based bidirectional long short term memory graph convolutional neural network for human motor imagery recognition. *Frontiers in Bioengineering and Biotechnology*, 9:706229, 2022.

[18] **Shuyue Jia**. EEG-DL: A deep learning library for EEG tasks classification. https://github.com/SuperBruceJia/EEG-DL, 2026.